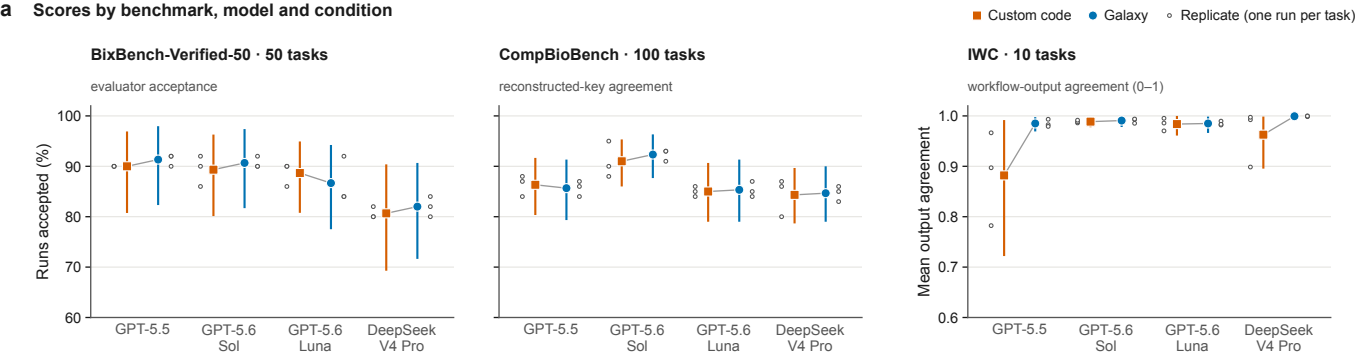
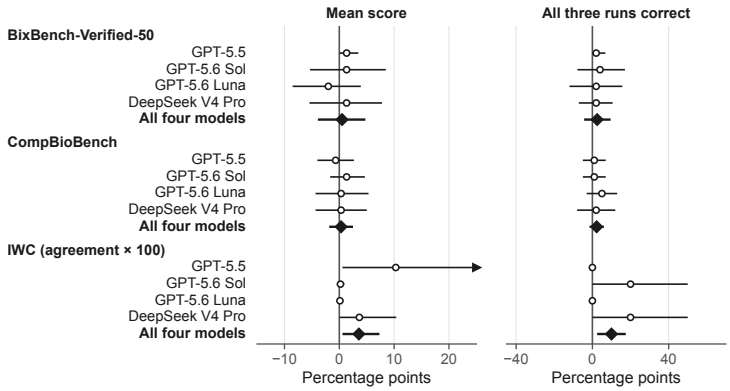
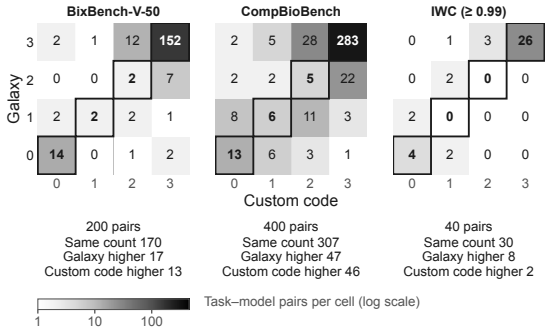
## a Scores by benchmark, model and condition




## b Paired differences: Galaxy minus custom code

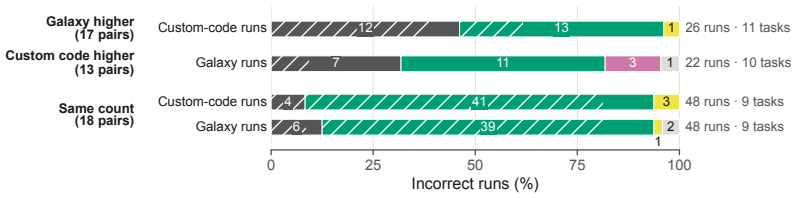



## c Correct runs of three, same task and model




## d Why the conditions disagree (BixBench-Verified-50)

Incorrect runs by primary cause (AI-assisted audit), for task–model pairs whose conditions differ or agree
An independent second rater agreed on the cause for 35 of 45 sampled runs (κ = 0.56)




### Traced case: bix-45-q1 (relative composition variability)

| | Custom code | Galaxy |
|---|---|---|
| GPT-5.5 | ○ ○ ○ | ○ ○ ○ |
| GPT-5.6 Sol | ● ● ● | ○ ○ ○ |
| GPT-5.6 Luna | ● ○ ● | ○ ○ ○ |
| DeepSeek V4 Pro | ● ● ● | ○ ○ ○ |
| | 8 of 12 correct | 0 of 12 correct |

Successful custom-code runs matched the PhyKIT version to the date of the input archive. The Galaxy PhyKIT wrapper, like current PhyKIT, returned a different value, and the wrapper version was not shown to the agent; the reference encodes the older value.

● correct ○ incorrect (one run each)